

CV 1주차

pixelight open cv 기초와 open cv 활용

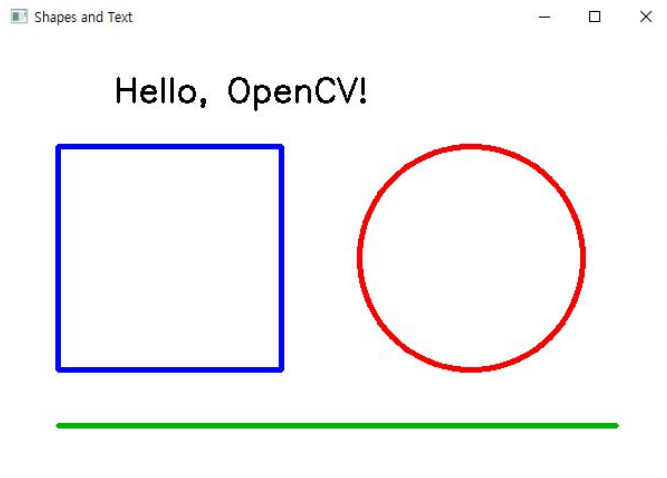
OpenCV란?

- 컴퓨터 비전에 사용되는 C++ 기반 오픈소스 라이브러리



OpenCV로 무엇을 할 수 있나?

- 이미지의 크기와 색상을 바꾸고, 도형과 글자를 그릴 수 있다.
- 사진이나 영상에서 윤곽선과 특징을 검출할 수 있다.
- 카메라 영상을 실시간으로 처리하고, 계산 결과를 시각화할 수 있다



OpenCV-Python

- OpenCV-Python은 파이썬에서 OpenCV의 기능을 사용할 수 있는 패키지
- `cv2` 모듈을 불러와 간단한 코드로 이미지 처리와 시각화를 구현 가능

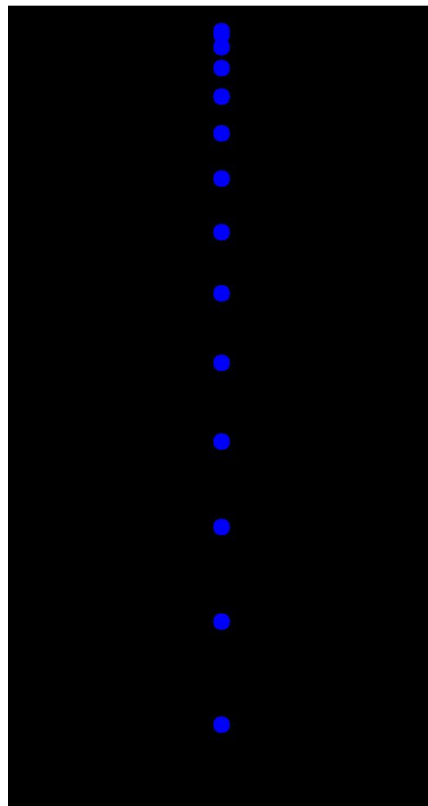


OpenCV를 이용한 자유 낙하 시각화

```
import cv2
from math import *
import numpy as np

cv2.namedWindow('Free Fall')

width = 512
height = 960
img = np.zeros((height,width,3), np.uint8)
time = ypos = 0
while (True):
    if (ypos+30) < height:
        cv2.circle(img, (256, 30+ypos), 10, (255, 0, 0), -1)
        time +=1
        ypos = int((9.8 * time**2)/2)
        print(time,':', ypos)
        cv2.imshow('Clock', img)
        if cv2.waitKey(100) >= 0:
            break
cv2.destroyAllWindows('Free Fall')
```



OpenCV를 이용한 피보나치 수열 시각화

```
import cv2
from math import *
import numpy as np
cv2.namedWindow('Pibonacci graphic')

width = 960
height = 960
img = np.zeros((height,width,3), np.uint8)
init_posx=480
init_posy=480
drawing_unit = 5
idx, a, b = 0, 0, 1

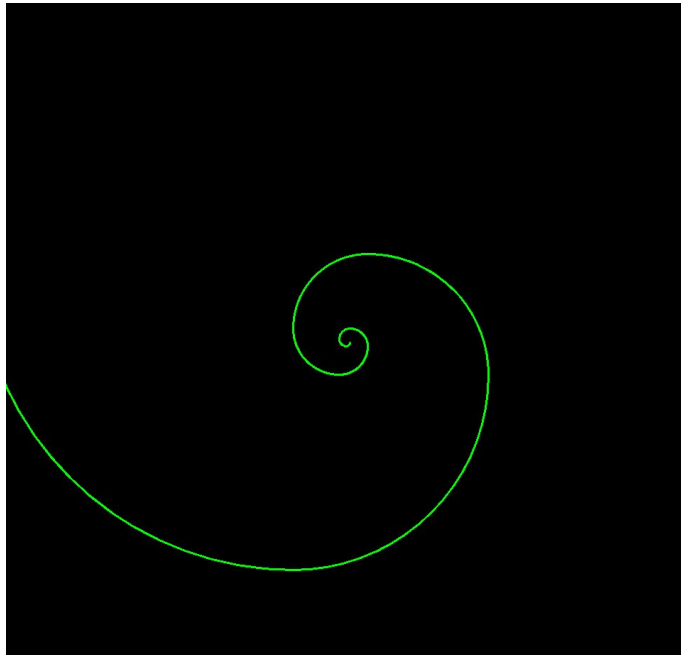
while a<70:
    cv2.imshow('Pibonacci graphicn', img)
    a, b = b, a + b
    Begin_Ang = int(90*(idx%4))
    End_Ang = int(90*(idx%4+1))
    if Begin_Ang == 0:
        init_posx -= (b - a) * drawing_unit
    elif Begin_Ang == 90:
        init_posy -= (b - a) * drawing_unit
    elif Begin_Ang == 180:
        init_posx += (b - a) * drawing_unit
    elif Begin_Ang == 270:
        init_posy += (b - a) * drawing_unit

    print(idx,'turn : ', 'Pibonacci number -', b)

    cv2.ellipse(img, (init_posx, init_posy), (b*drawing_unit,b*drawing_unit), 0, Begin_Ang, End_Ang, (0,255,0), 2)

    idx+=1

while (True):
    if cv2.waitKey(10) >= 0:
        break
cv2.destroyAllWindows('Pibonacci graphic')
```



OpenCV를 이용한 포물선운동 시각화

```
import cv2
from math import *
import numpy as np

init_Vel = float(input("초기 속도를 입력하세요 : "))
init_Ang = float(input("초기 각도를 입력하세요 : "))
cv2.namedWindow('Parabolic Motion')

width = 960
height = 960
img = np.zeros((height, width, 3), np.uint8)

time = xpos = ypos = 0
init_posx = 30
init_posy = 250
Vel_x = int(init_Vel*cos(init_Ang*pi/180.0))
Vel_y = int(-1.0*init_Vel*sin(init_Ang*pi/180.0))

while(True):
    if(ypos+30) < height:
        cv2.circle(img, (init_posx+xpos, init_posy+ypos), 10, (255, 0, 0), -1)
        time += 0.2
        Vel_y = int(Vel_y + 9.8 * time)

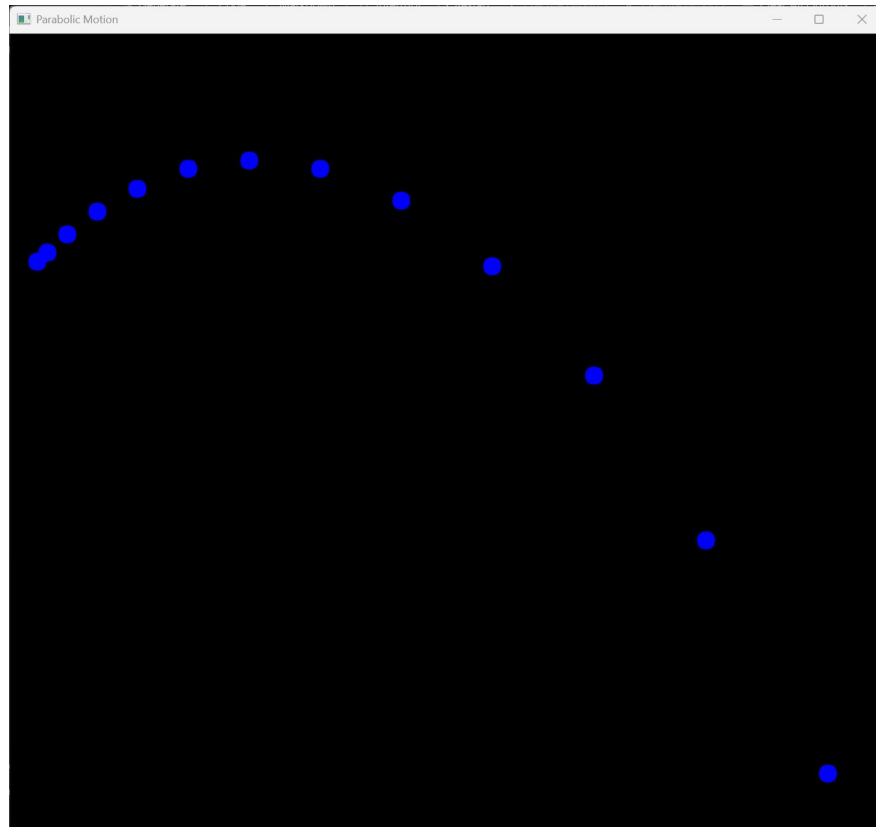
        xpos = int(xpos + Vel_x*time)
        ypos = ypos + int(Vel_y*time) + int((9.8 * time**2)/2)
        print(time, ':', xpos, ypos)

    cv2.imshow('Parabolic Motion', img)

    if cv2.waitKey(100) >= 0:
        break

cv2.destroyAllWindows()

✓ 4.5s
```



OpenCV를 이용하여 신호등 만들기

```
import cv2
import numpy as np
import time

width, height = 420, 620
window_name = "Traffic Light"

# Durations for red, green, and yellow lights, in seconds.
phases = [("RED", 4), ("GREEN", 4), ("YELLOW", 2)]
light_colors = {
    "RED": (0, 0, 255),
    "GREEN": (0, 255, 0),
    "YELLOW": (255, 255, 0),
}
dim_color = (55, 55, 55)

cv2.namedWindow(window_name)
start_time = time.monotonic()

while True:
    elapsed = time.monotonic() - start_time
    cycle_duration = sum(duration for _, duration in phases)
    cycle_time = elapsed % cycle_duration

    for phase_name, duration in phases:
        if cycle_time < duration:
            remaining = int(duration - cycle_time) + 1
            break
        cycle_time -= duration

    frame = np.full((height, width, 3), (235, 235, 235), dtype=np.uint8)
    cv2.rectangle(frame, (125, 45), (295, 465), (35, 35, -1))
    cv2.rectangle(frame, (195, 465), (225, 600), (70, 70, 70), -1)

    light_positions = {"RED": (210, 120), "YELLOW": (210, 255), "GREEN": (210, 390)}
    for name, center in light_positions.items():
        color = light_colors[name] if name == phase_name else dim_color
        cv2.circle(frame, center, 48, color, -1)
        cv2.circle(frame, center, 48, (190, 190, 190), 3)

    cv2.putText(frame, phase_name, (145, 535), cv2.FONT_HERSHEY_SIMPLEX, 1.0,
                (20, 20, 20), 2, cv2.LINE_AA)
    cv2.putText(frame, f"Next change: {remaining}s", (105, 575),
                cv2.FONT_HERSHEY_SIMPLEX, 0.65, (20, 20, 20), 2, cv2.LINE_AA)
    cv2.imshow(window_name, frame)

    key = cv2.waitKey(30) & 0xFF
    if key == 27 or key == ord("q"):
        break

cv2.destroyAllWindows()
```

